

TRIBHUVAN UNIVERSITY — INSTITUTE OF ENGINEERING

EXAMINATION CONTROL DIVISION

PAST QUESTION COMPILATION

SUBJECT: Software Engineering (CT 601)

YEAR / PART: III / I

PROGRAMME: Computer Engineering (BCT)

FULL MARKS: 80

SESSION 1: 2083 Baishakh (Back)

1. What is software crisis? Justify this statement "Software doesn't wear out". [3 + 4]
2. Why do we need to perform requirement analysis and refinement during requirement engineering process? Explain the spiral model of software development. Why is it preferred in high-risk projects? [3 + 5]
3. Why do we need to do feasibility study before accepting any project? Differentiate between functional and non-functional requirement with example. [3 + 4]
4. Design Level 0 and Level 1 DFD for the case study given below: [6] Case Study: Suppose a buses garage repairs wants to develop an automation system that helps to maintain their records. In the garage, buses come for repairs. A mechanic and helper perform the repair, record the reason for the repair and record the total cost of all parts used on a shop repair order. Information on labor, parts and repair outcome is used for billing by the accounting department, parts monitoring by the inventory management computer system and a performance review by the supervisor.
5. Draw the use case diagram for the following system: [4] Case Study: In an online food delivery system where the customer order food and make payment from the Fonepay by providing the necessary information. Food delivery system validates the payment and processes the order to deliver. The delivery person takes the order and ship to the customer address. The customer receipts the order and acknowledges the food ordering system.
6. What is an architectural design? Why architecture is important to drive software development? Explain 2-tier and 3-tier architecture with example. [1 + 3 + 4]
7. Why continuous monitoring and controlling is required in Real time System? Differentiate between hard and soft real time system. [2 + 2]
8. What are the benefits and limitations of software reuse? [4]
9. Explain provided and required interfaces in a software component with example. [4]
10. Differentiate between verification and validation with examples. What type of system are validated using the formal methods and why? [3 + 3]
11. Why end-user can't perform white-box testing? Compare integration testing and incremental integration testing. [2 + 2]

12. Calculate COCOMO effort, development time in calendar month, average staffing for a project of application program that is estimated to be 55,000 lines of code. [4]
13. What is the significance of "the vital few" in statistical SQA? How the software quality assurance help to enhance quality of a software? Why do we prefer lower value of Fan-out for any function in a program as software metric? Explain. [2 + 3 + 2]
14. Describe the staged CMMI model with levels and key process areas. Differentiate version and variants in the context of configuration management. [5 + 2]

SESSION 2: 2082 Bhadra (Regular)

1. Explain how modern software engineering practices helped overcome the software crisis. Justify this statement "Software doesn't wear out like hardware components". [3 + 4]
2. Why do we need to do feasibility study before accepting any project? Prepare functional requirements based on the given scenario: [2 + 4] Case Study: A movie theatre wants to develop a system for online ticket booking. The system should allow customers to search for movie show times, book tickets, and make payments. Admin should be able to manage movie listings, update show times, handle ticket sales, and generate sales reports.
3. Explain fundamental activities of the software process. You are leading a software development team for developing an e-commerce site. You choose the incremental model for software development. How do you implement the incremental model? [3 + 4]
4. Design Level-0 and Level-1 DFD for the case study given below: [5] Case Study: You are tasked with designing a software system for managing examinations in Tribhuvan University. The Examination Management System (EMS) should automate and support the examination functionalities. Students log in to the system and register for the exams they are eligible for. The Exam Controller uploads and manages the exam timetable (date, time, subject, room number). Once registered, students should be able to download his/her admit card. After exams, the subject teachers upload the marks, and students can view their results. The system should automatically calculate grades based on predefined criteria. The system must store student data, subject details, schedules, results, and grades securely.
5. Prepare use case diagram for the given scenario: [4] Case Study: An event management company wants to develop a system to organize and manage various events such as conferences, weddings, and concerts. The system should allow users to view and register for events, select tickets, and make payments. Admin should be able to create and manage event details, track registrations, handle user inquiries, and generate attendance reports.
6. Differentiate between thin client and thick client model. Describe layered architecture for software. [3 + 4]
7. What is real-time system? Explain data acquisition system. [1 + 3]
8. What are the benefits and risks of software reuse? What factors must be considered during reuse planning? [4 + 2]
9. What are provided and required interfaces in a software component? Explain with the help of a suitable example. [4]
10. What is a software inspection process? What are the faults that can be uncovered by software inspection? What type of system are validated using the formal methods and why? [4 + 3]

11. What is equivalence partition? How it helps in software testing? Compare Alpha and Beta testing. [2 + 2]
12. The size of a software is estimated to be 300 KLOC. Using COCOMO model, calculate: (i) Effort (PM) (ii) Development Time (months) and (iii) Average Team Size. [4]
13. What is the significance of "the vital few" in statistical SQA? How the Formal Technical Review help to enhance quality of a product? Why do we prefer lower value of Fan-In for any function in a program as software metric? Explain. [2 + 3 + 2]
14. What is release problem? Describe the Staged CMMI model with levels and key process areas. Differentiate Version and Variants in the context of configuration management. [1 + 5 + 2]

SESSION 3: 2082 Baishakh (Back)

1. What is the importance of SRS document in software development? List and explain functional (any two) and non-functional requirement of an airlines reservation system. [3 + 6]
2. What are the benefits of CASE tools in software engineering? Justify this statement "Software doesn't wear out". [3 + 5]
3. In what types of project, the prototype process model can be used? Explain with an example. [5]
4. Why architectural design is important in software engineering? Explain client-server architecture with suitable example. [2 + 5]
5. Draw level-0 and level-1 data flow diagram of "Student Attendance Management System". Assume all the necessary and required specifications on your own and state them clearly. [6]
6. Draw use case diagram for the online food ordering system, where user can order food and make payment through the app by providing necessary information for food delivery. Make necessary assumptions if necessary. [5]
7. Differentiate between aperiodic and periodic stimulus in real time system. What is the use of data acquisition system (DAS)? [2 + 3]
8. "Component based software engineering is a reuse-based approach to defining and implementing loosely coupled components into a system." Justify this statement. [4]
9. What are the pros and cons of software reuse? What are the necessary factors that need to be considered while software reuse planning? Explain. [5 + 3]
10. Differentiate between verification and validation. What are the faults that can be uncovered by software inspection? [3 + 2]
11. Differentiate between black-box testing and white-box testing. An EMIS System has 30 user input (weighting factor = 3), 75 user output (weighting factor = 7), 40 user inquiries (weighting factor = 6), 7 user files (weighting factor = 7) and 5 external interfaces (weighting factor = 10), using functional point analysis compute FP for the system. Assume complexity factors are high for all. [3 + 5]
12. Explain different levels of CMMI model to evaluate the maturity of a software development. [5]
13. Define version, variants with respect to configuration management. Describe change management process. [2 + 3]

SESSION 4: 2081 Bhadra (Regular)

1. Why it is so difficult to gain a clear understanding of what the customer wants? Explain functional (order Food and make Payment) and non-functional (All) of an online food ordering system. [3 + 6]
2. What are the characteristics of good software? Justify this statement "Software doesn't wear out". [2 + 5]
3. Explain Incremental Model for software development with its pros and cons. [6]
4. What is an architectural design? How is the modular decomposition concept practiced in the system design process? [1 + 5]
5. Design Level-0 and Level-1 DFD for the case study given below: [7] Case Study:
Suppose a travel agency wants an Airline Ticketing System to be developed for the office so that user can easily book flight tickets from anywhere. The customer needs to create an account if he/she is new to the system. Then the customer enters destination and data for the flight. After that, the system displays the available airlines for the same along with route or available time which is provided by the airlines company. Now the customer selects the airline which he/she finds appropriate where he/she can either book the ticket or confirm the ticket. The customer pays the ticket charge via transferring the amount to the agency's bank account directly. The customer has to provide the valid email address to get the notification of booking or ticket confirm.
6. Draw use case diagram for the online Jyotis appointment booking App, where user can take appointment and make payment through the App by providing necessary information. After user validation the Astrologer can provide the online consultation to the user. [5]
7. Differentiate between hard real time system and soft-real time system. What is the use of real time operating system (RTOS)? [2 + 2]
8. "Component based software engineering is a reuse-based approach to defining and implementing loosely coupled components into a system." Justify this statement. [4]
9. What are the benefits of software reuse? What are the necessary factors that need to be considered while software reuse planning? Explain. [3 + 3]
10. Explain Software Inspection Process. What are the faults that can be uncovered by software Inspection? [4 + 2]
11. Differentiate between alpha testing and beta testing in system testing. A MIS System has 60 user input (weighting factor = 4), 75 user output (weighting factor = 5), 40 user inquiries (weighting factor = 4), 7 user files (weighting factor = 10) and 5 external interfaces (weighting factor = 7), using functional point analysis compute FP for the system. Assume complexity factors are average for all. [2 + 5]
12. What is the importance of the "Vital few" in statistical SQA? Explain. How CMMI model is used to evaluate the maturity of a software development? Explain. [2 + 5]
13. Define version, variants & release with respect to configuration management. Describe change management process in software engineering. [3 + 3]